

Solutions to Practice Exercise on the Ricardian Model

Problem 1: Comparative Advantage and Opportunity Cost

Two countries, **A** and **B**, produce two goods: **Wheat** and **Cloth**. The labor productivity (output per worker per hour) is given below:

Country	Wheat (units per hour)	Cloth (units per hour)
A	10	5
B	4	2

Solution:

(a) Opportunity cost of wheat in country A: $\frac{5}{10} = 0.5$ units of cloth.

Opportunity cost of wheat in country B: $\frac{2}{4} = 0.5$ units of cloth.

(b) Opportunity cost of cloth in country A: $\frac{10}{5} = 2$ units of wheat.

Opportunity cost of cloth in country B: $\frac{4}{2} = 2$ units of wheat.

(c) Since the opportunity costs are the same, neither country has a comparative advantage.

Problem 2: Gains from Specialization and Trade

Consider two countries, **X** and **Y**, producing **Cars** and **Computers**. The total labor force is **1,000 workers** in each country. The labor productivity is:

Country	Cars per worker per year	Computers per worker per year
X	4	8
Y	2	6

Solution:

(a) Opportunity cost of 1 car in X: $\frac{8}{4} = 2$ computers.

Opportunity cost of 1 car in Y: $\frac{6}{2} = 3$ computers.

(b) X specializes in cars, Y specializes in computers.

Total production: $1000 \times 4 = 4000$ cars and $1000 \times 6 = 6000$ computers.

(c) Both countries benefit by trading at a rate between 2 and 3 computers per car.

Problem 3: Terms of Trade and Trade Gains

Two countries, **Brazil** and **Canada**, produce **Coffee** and **Timber**. The labor productivity is as follows:

Country	Coffee (tons per worker)	Timber (tons per worker)
Brazil	12	4
Canada	8	6

Solution:

(a) Opportunity cost of 1 ton of coffee:

Brazil: $\frac{4}{12} = 0.33$ tons of timber.

Canada: $\frac{6}{8} = 0.75$ tons of timber.

(b) Since Brazil has a lower opportunity cost for coffee, it should specialize in coffee, and Canada in timber.

At the trade rate of 1 coffee = 0.5 timber, both countries gain as it is between their opportunity costs.

Problem 4: Effect of Technological Change on Trade Patterns

Initially, two countries, **India** and **Germany**, produce **Steel** and **Textiles**. The productivity levels are:

Country	Steel (tons per worker)	Textiles (meters per worker)
India	5	20
Germany	10	30

Solution:

(a) Opportunity cost of 1 ton of steel:

India: $\frac{20}{5} = 4$ meters of textiles.

Germany: $\frac{30}{10} = 3$ meters of textiles.

Germany has a comparative advantage in steel, and India in textiles.

(b) If Germany doubles its steel productivity, it can now produce 20 tons per worker.

New opportunity cost of steel in Germany: $\frac{30}{20} = 1.5$ meters of textiles.

Germany's comparative advantage in steel increases.

(c) Germany will export more steel, and India will export more textiles. The gains from trade will increase.